Automated Kidney Lesion Classification from CT Images Using Convolutional Neural Networks with Explainability

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A*BSTRACT*
Nowadays, due to people's lifestyle habits such as unhealthy diets, less exercise, and stress are leading to more kidney-related issues. There are three types of kidney lesions that can be seen on CT scans: cysts, stones and tumors. These conditions can have different outcomes depending on how early they are diagnosed. Looking at CT scans to find these problems is a lot of work. Most importantly it depends on the person looking at the scans. This is especially true in places where they do not have a lot of resources.
This article is about Automated Kidney Lesion Classification from CT Images which can automatically classify kidney lesions from CT images using Convolutional Neural Networks to tell if the kidney is normal or if there is a cyst, stone or tumor. What it does exactly is it takes your uploaded CT scanned image of kidney then processes it and gives you the analysis of it with the results. It makes a map that highlights the problem areas with the help of Grad-CAM. It also shows the accuracy using confidence bar. Finally generates a detailed report of the patient and also aims to answer any queries with the help of AI chat assistant.

Keywords: *Kidney Lesion Classification, CNN, Transfer Learning, Grad-CAM, Explainable AI, CT Imaging, ResNet50, Clinical Decision Support*

PROBLEM STATEMENT
Automated kidney lesion classification has seen a lot of progress over the past few years. However, there are still many issues which may result in serious problems.
One of the problems is that kidney lesions are often very small and hard to see in CT scans. Hence, it makes tough for doctors to treat them correctly. Kidney lesions can also look a lot alike which makes it hard to tell them apart.
There is also no explanation of how kidney lesion classification systems work. In places where resources are limited it is even harder to manage a lot of patients and get it right. If we make a mistake with kidney lesion classification it is not a technical issue. This means that the patient will not get the treatment, for their kidney lesion.
This project is trying to solve these kidney lesion classification problems.

I.INTRODUCTION:
Kidney disease is a health problem worldwide affecting over about 850 million peopleThe three main types of kidney lesions that can be seen on CT scans are cysts, stones and tumors. These conditions can have different outcomes depending on how early they are diagnosed. For example if kidney cancer is diagnosed early the five-year survival rate is over 93%. However if it is not diagnosed until later the survival rate drops to below 12%. This highlights the importance of timely diagnosis.

CT scans are the way that kidney lesions are diagnosed. However interpreting CT scans is an time-consuming task that requires a lot of expertise. Primarily radiologists need to have an in depth knowledge about various conditions in addition they need to make the decision on basis of what they see.
This can be challenging in cases where the lesions are small or not clearly visible. In healthcare settings with the increasing number of patients it becomes challenging to keep up the demands.

The use of deep learning has increased in past few years for analyzing medical images. Convolutional Neural Networks (CNNs) are a type of deep learning model which can be used to classify images. These CNNs work well as the look at pictures and figure out what is in them. They do this by checking out the dots of color in a picture called pixels and using that to guess what the picture shows to make predictions.

The main goal of this project was to create a system that can be used to classify kidney lesions from CT images using Convolutional Neural Networks.
This system is designed to be easy to understand so it can tell us how it makes its predictions.
This system takes care of everything from uploading the image to giving us the results. Here is what it does:

- Compares four models with different architectures to see which one is the best.
- Grad-cam to explain every prediction it makes
- It helps figure out the level of risk based on how confident it is with the prediction.
- It generates reports and also provides more knowledge about the condition with AI chat assistant.

II.LITERATURE REVIEW

Nowadays there has a been a significant rise of AI based models in healthcare sector. Deep learning for kidney lesion classification has become more popular over the few years.It’s because of their potential to improve the precision of and help with faster more reliable results for patients.
There have been several studies on kidney lesion classification using deep learning.

Subedi et al [5] compared ResNet50, VGG16 and a Vision Transformer on the same Kaggle CT dataset.
The Vision Transformer got a high accuracy of 99.64% on a 90:10 split.However, it does not explain how it makes theses decisions and lacks support to back it.

Pande and Agarwal [6] used YOLOv8n-cls. and got an accuracy of 82.52%. YOLO is fast but it does not extract features in depth.
Mehedi and others [7] showed that adding segmentation with UNet and VGG16 increases accuracy to 99.48%.
This method requires pixel-level annotation but Clinical datasets do not have this level of detail.

In transfer learning Pimpalkar et al.[10] Got an accuracy of 99.96%, with a changed InceptionV3 model and data augmentation.However,it can be difficult to get this result if you do not have the same setup. The model might only work well with their specific dataset and not on different datsets. Pimpalkar et al. used transfer learning approaches to get this result.

Majid et al. [11] worked with DenseNet-121 and got great results achieving 98.22% accuracy. Along with that, Ahmed et al. [12] used a different approach by using VGG16 model with Grad-CAM on X-ray images and achieved 97.41% accuracy.
These works are important because it helps us understand why the DenseNet-121 and VGG16 models make the decisions they do. The DenseNet-121 and VGG16 models are really good, at what they do and understanding how they work is crucial.
Despite having high accuracy there is a big gap, in current research.

Most systems focus on accuracy. Ignore other key aspects. These aspects include explainability, uncertainty and clinical support.
Many studies also do not validate their results on datasets.
We need to be more careful when we see high accuracy on a particular dataset. Keeping this in mind, our system is not just about performance numbers.

The kidney lesion classification system is designed to enhance knowledge of people and also doctors who are still new in the field to help them get wider picture of the condition.
It is made to help them. It tries to give them a more complete and practical solution, for kidney lesion classification.

III.OBJECTIVES

1. Build a system that looks at pictures of kidneys and figures out if the kidneys are normal or if they have something with them like a cyst or a stone or a tumor.
2. Compare four different models which are, a custom model, MobileNetV2, EfficientNetB0 and ResNet50 to see which one works best for our dataset.
3. Use Grad-CAM to create pictures that show doctors which parts of the kidney image are affected and the computer focused on to make its decision.
4. Create a dashboard that tells doctors how sure the computer is about its decision using the confidence bars.
5. Build a system that helps doctors and patients with a treatment plan based on the illness they are dealing with.
6. Check if the system actually works well by looking at curacy, precision, recall, F1-score and confusion matrix analysis.

IV.CONTRIBUTIONS

1. We compared four different models to see which one works best with the same dataset.
2. We used Grad-CAM to highlight the areas on a scan that a doctor would look at so we know the system is making decisions based on medical evidence.
3. We sort patients by risk level based on how sure we are using the confidence score.
4. We built an AI assistant which can be helpful for doctors decide what to do based on what the system predicts and also help the patients to learn more about their condition.
5. We created a way to download the report of the prediction that is easy for patients and doctors to use and understand.

V.METHODOLOGY

A. Dataset

The project uses a CT Kidney Dataset that was first shared on Kaggle by Islam et al. [16]. This CT Kidney Dataset originally had 12,446 images. However, the dataset for this project used a smaller set with a better balance of images.
The CT Kidney Dataset used for the project has 5,708 CT images that are divided into four groups:
Cyst: 1,500 images
Normal: 1,500 images
Tumor: 1,500 images
Stone: 1,208 images

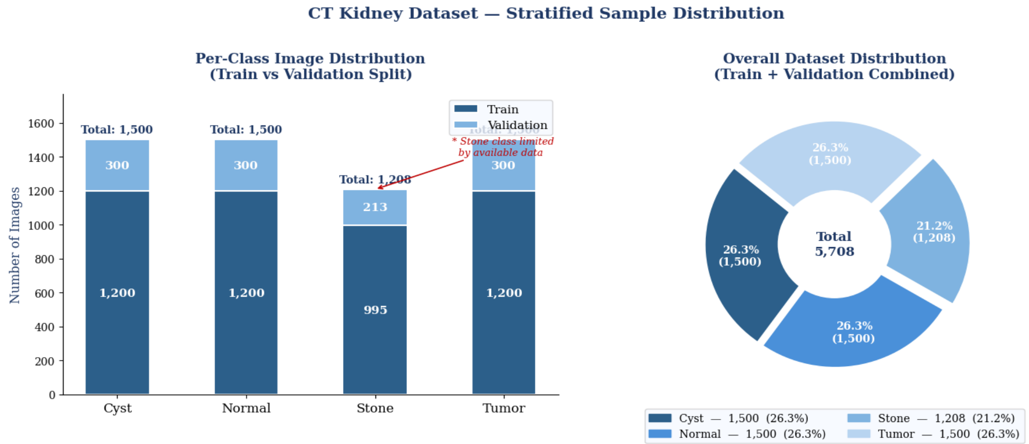


Fig. 1. *Distribution of dataset classes (total = 5,708 images). The* ***Stone*** *class has the fewest samples (11.1%), so we handled this imbalance using class weights and some extra sampling during training.*

The CT Kidney Dataset was split into two parts: one for training and one for validation. It was ensured that the number of images in each group was similar in both parts. Out of all the images in the CT Kidney Dataset:
4,559 images were used to train the model
1,149 images were used to validate the model

For most of the groups, such as the Cyst group, the Normal group and the Tumor group we used 1,200 images to train the model and 300 images to validate the model. The Stone group had images so the project team used 995 images to train the model and 213 images to validate the model.

The validation part of the CT Kidney Dataset was kept separate, from the training part. We did not change or add to the validation images so the results show how well the model really works with images it has not seen before.

B. Preprocessing and Augmentation

All the images were made to be the size, which is 224 by 224 pixels. The color of the pixels in the images was also made to be the same by dividing the numbers by 255 so they are all between 0 and 1.
Few changes were made on the images to make the model better at recognizing kidneys.

It was done ImageDataGenerator from Keras. The images were rotated a bit up to 15 degrees to make the model see the kidneys from angles. They were also made a bit wider or narrower up to 5 percent to make the model better at recognizing kidneys that're not perfect. Sometimes the image was flipped horizontally like it was mirrored, to make the model see the kidneys from the side.

The images were also made a bit smaller up to 10 percent to make the model better at recognizing kidneys of different sizes.
The empty spaces in the images were filled with the color of the pixel next to it which is the color of the kidney.

The model was not shown any images that were flipped down because those images would not show a kidney.
We only want the model to recognize pictures of kidneys.
There was a little problem with the data because the Stone images were 11.1% of all the images.

More Stone images were added to the data so the model can see more examples of kidneys, with a Stone.
The model was also told to give weight to the types of images that were not as common like the Stone images so it can recognize them better.

Augmentation Parameter	Value / Configuration
Resize	224 × 224 px (bilinear)
Normalization	1/255.0 → [0, 1]
Rotation	±15°
Width / height shift	±5%
Zoom	±10%
Horizontal flip	Enabled
Vertical flip	Disabled (anatomical constraint)
Fill mode	Nearest neighbor

Table I. Training augmentation parameters

VI.SYSTEM ARCHITECTURE

The system is made up of a series of steps that happen one after the other. First a CT scan is uploaded to the system. Then it is cleaned up and prepared for use. After that it is passed through a trained CNN classifier. The CNN classifier is then used for Grad-CAM analysis. This analysis helps the system figure out which parts of the picture are the important by highlighting the important regions.

Based on this analysis confidence score is calculated. The confidence score is used to decide what next steps should be taken. Finally, the patient can download a detailed pdf report. After this if the patient has any queries regarding their condition, they can use the AI assistant to clear their doubts

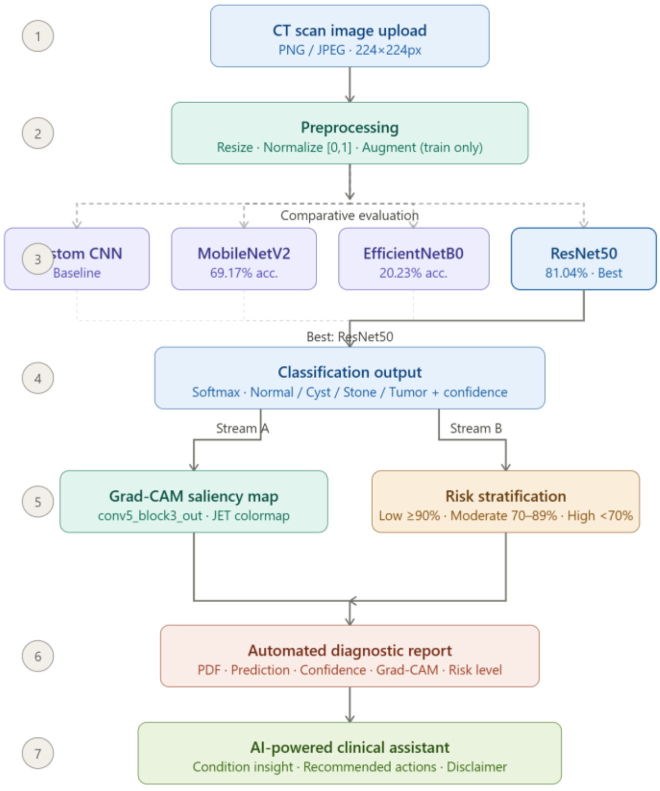


Fig. 2. Flow of the system from start-to-end.

A. Model Architectures

Four different models were built and tested under the same training conditions to make a fair comparison.

1) Custom CNN (Baseline)

The model has three convolutional blocks (Conv2D + BatchNorm + MaxPool), followed by global average pooling. Then it has two dense layers (512 and 128 units), with a dropout of 0.5, and ends with a four-class softmax layer. It was trained from scratch without using any pre-trained weights to establish a baseline performance.

2) MobileNetV2

MobileNetV2 is a good model because it is efficient and uses less computation power. For this project we used MobileNetV2 with weights it already knew from ImageNet. The original final layer was replaced with a dense layer of 128 units and a dropout of 0.3. At first, we did not change most of the layers in MobileNetV2 so the new layers could learn properly. After that, deeper layers (from layer 100 onward) were unfrozen and fine-tuned to improve performance.

3) EfficientNetB0

This model improves performance by scaling depth, width, and image size. It also uses MBConv blocks with squeeze-and-excitation. It also uses the same classification head as MobileNetV2.However the model did not work well when it was used to look at CT scans. The model had trouble when it went from looking at pictures to looking at CT scans as the model was sensitive, to this change from natural images.

4) ResNet50

This model uses residual (skip) connections, which allow it to go deeper (48 layers) without training issues like vanishing gradients. It has around 25.6 million parameters organized into four stages.
The custom classification head includes global average pooling, followed by a 256-unit dense layer, a dropout of 0.3, and a four-class softmax output.
Overall, this model gave the best performance across all evaluation metrics in comparison to other models.

B. Transfer Learning Strategy

Each of the three transfer learning models was trained in two steps. This helped make use of the pre-trained knowledge while adapting to CT scan data.

In Phase 1 the main model, also called the backbone was completely frozen. Only the new classification head was trained for 10 epochs. A small learning rate of 0.001 was used which helps the new layers learn properly without changing the -trained features.

In Phase 2 a few layers, at the end of the model were unfrozen. The whole network was then trained together for another 20 epochs. A smaller learning rate of 0.0001 was used. This smaller learning rate
helped prevent damaging the patterns that the model learned from ImageNet data.

Hyperparameter	Custom CNN	MobileNetV2	EfficientNetB0	ResNet50
Optimizer	Adam	Adam	Adam	Adam
LR Phase 1	0.001	0.001	0.001	0.001
LR Phase 2	N/A	0.0001	0.0001	0.0001
Batch size	32	32	32	32
Epochs (Ph. 1)	25	10	10	10
Epochs (Ph. 2)	N/A	20	20	20
Dropout	0.5	0.3	0.3	0.3
Dense units	512, 128	128	128	256

Table II. Hyperparameter configuration for all four models.

C. Grad-CAM Explainability

Grad-CAM was used to understand how the model makes its predictions. It generates heatmaps that highlight the important areas in the image that influenced the result.

The method looks at the last convolutional layer and calculates how much each feature contributes to a specific class. This contribution is calculated as:

$$\alpha_k^c = \frac{1}{Z}\sum_i\sum_j \frac{\partial y^c}{\partial A_{ij}^k}$$

These values are then combined to form the heatmap:

$$\text{Grad-CAM} = \text{ReLU}\left(\sum_k \alpha_k^c A^k\right)$$

The heatmap is then placed over the original image, where warm colors (red, yellow) show important regions and cool colors (blue) show less important ones. Only positive contributions are considered.

For ResNet50, Grad-CAM was applied to the **'conv5_block3_out'** layer.

D. Confidence-Aware Risk Stratification

The model produces a probability score for each of the four respective classes using the softmax outputs. The maximum score was used as the prediction confidence score.

Instead of just showing this number to the user, I created a simple three-level risk grouping system that converts the model's confidence into a clear recommendation on what action to take next.

Confidence	Risk Level	Clinical Action
≥ 90%	**Low Risk**	Proceed; routine expert advised.
70–89%	**Moderate Risk**	expert review is strongly recommended before taking action.
< 70%	**High / Uncertain**	An expert review is required, and decisions should not be made based only on the model output.

Table III. Risk level framework based on confidence score.

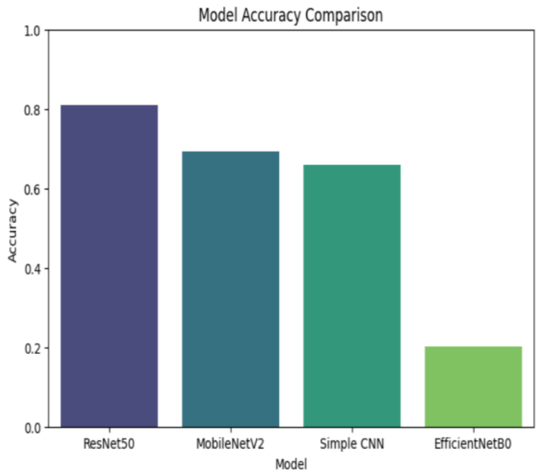
E. AI Assistant and Diagnostic Reports

We have an AI chat assistant through which the patient can clear any doubts that they have. It mentions possible symptoms that should be paid attention to, along with suggesting next steps, it also gives basic lifestyle advice. I want to make it clear that the system is only meant to support people and does not give any diagnosis to them. The advice it gives is based on medical rules.

After the system makes a prediction it automatically creates a two-page PDF report. This report has the condition the system predicted a score that shows how confident it is in its prediction with the level of risk, a Grad-CAM visualization probability for all classes, suggested next steps, and a disclaimer about the system's limitations.

VII.RESULTS

The performance of the models was evaluated using a validation set that had 2,489 images. Macro-averaged metrics were used for this. It means that the performance of the models was evaluated in a way that each class was given the equal importance. The performance of the models was evaluated in this way regardless of how samples each class had. This ensured fair evaluation of the models. It was especially important to evaluate the models in this way because the dataset of images that was used is not balanced.



Model	Accuracy	Precision	Recall	F1-Score
Custom CNN (Baseline)	65.99%	0.671	0.660	0.657
MobileNetV2	69.17%	0.879	0.692	0.718
EfficientNetB0	20.23%	0.058	0.202	0.090
ResNet50 (Best)	**81.04%**	**0.865**	**0.810**	**0.819**

Table IV. Validation performance of all four architectures (macro-averaged).

A.Model Comparison

The ResNet50 model did well compared to all the other models in every way. The ResNet50 model has skip connections which helps it learn more about the pictures making it better to find differences in texture and density between different kidney conditions. The other models were not as good at finding these details. The custom CNN model only got it right 65.99% of the time which shows that using models that have already been trained on lots of pictures, like the ones trained on ImageNet is a good idea even if you only change the last few layers of the ResNet50 model.

MobileNetV2 had a problem. It was really good at being precise and got the predictions right most of the time with a precision of 0.879 but it was not good at finding all the positive cases as its recall was only 0.692. This means that when MobileNetV2 said when it said a case was positive, it was mostly right, but it failed to detect a lot of true cases. In a hospital this can be a big issue because if you miss someone who is really sick it can cause serious harm to patient.

EfficientNetB0 gave the most surprising result with an accuracy of only 20.23%, which is really close to just guessing. This shows that EfficientNetB0 struggled to adapt from images to grayscale CT scans. EfficientNetB0 just could not do it. This highlights a point about EfficientNetB0 and other models that the models which do well on ImageNet may not always do well with medical imaging tasks, like CT scans.

B. Confusion Matrix Analysis

The results for each class from the confusion matrices, which are shown in Figures 3 to 6 give us details about the models. We can learn more about the models beyond looking at the numbers.

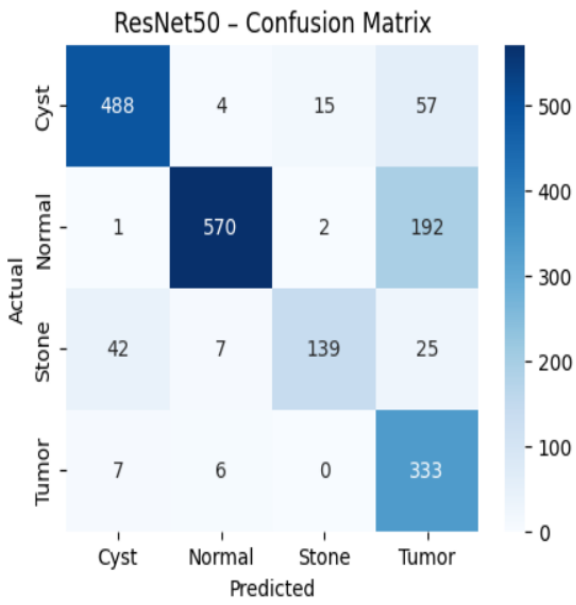


Fig. 3. ResNet50 confusion matrix.

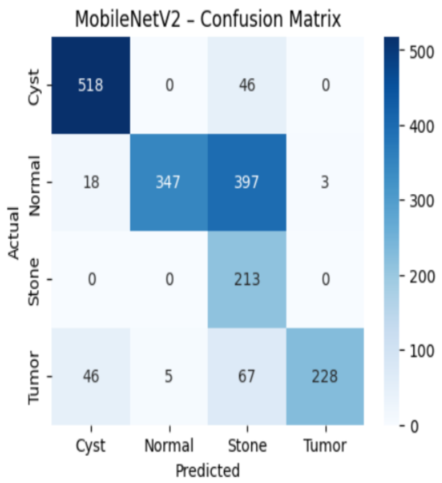


Fig. 4. MobileNetV2 confusion matrix.

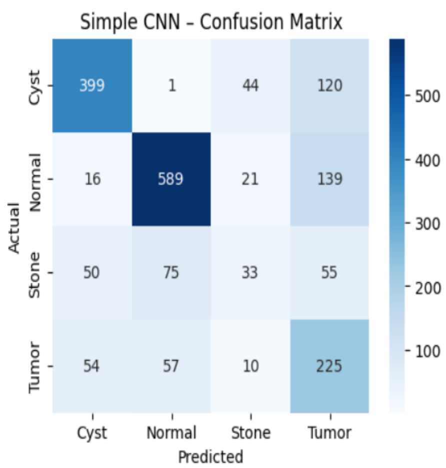


Fig. 5. Custom CNN confusion matrix.

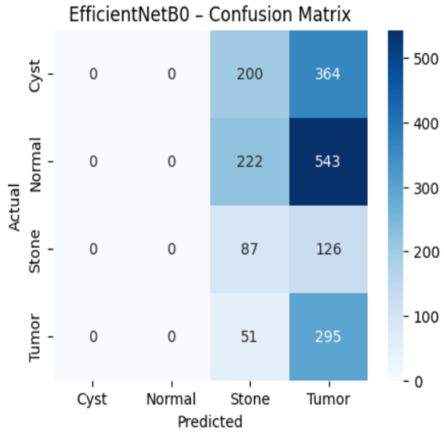


Fig. 6. EfficientNetB0 confusion matrix.

For ResNet50, the model correctly classified 488 Cyst, 570 Normal, 139 Stone, and 333 Tumor cases. The most biggest mistake was that 192 Normal cases were predicted as Tumor. While this is not the most ideal situation, it is still safer in a healthcare settings to flag a healthy case for review than to miss an actual tumor.

The Stone class did the worst with only 65.3% of cases detected. This might be because of the few numbers of stone images in the data and also because small stones look like blood vessels, in CT scans.

MobileNetV2 had a different kind of mistake. It did a good job with the Stone class by correctly classifying all 213 cases. However, it wrongly predicted Normal cases as Stone with just 397 out of 765 which is not safe.

When we look at the results of ResNet50 and MobileNetV2 they both have different strengths and weaknesses. This suggests that combining them (as an ensemble) could help balance their limitations and improve their overall performance.

The Custom CNN did not do well since it was trained from scratch, this makes sense because This is expected because it did not have any -learned features to build on.

EfficientNetB0 did the worst. Since, its confusion matrix shows that it mostly predicted only one class for all the inputs indicating it did not learn patterns.

C. Grad-CAM and Clinical Interface

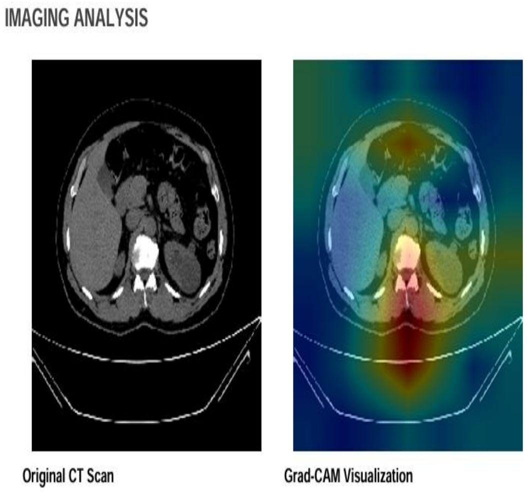
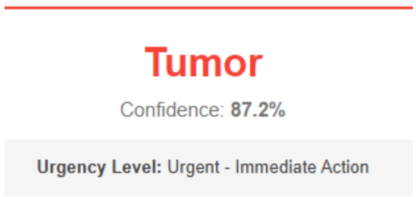
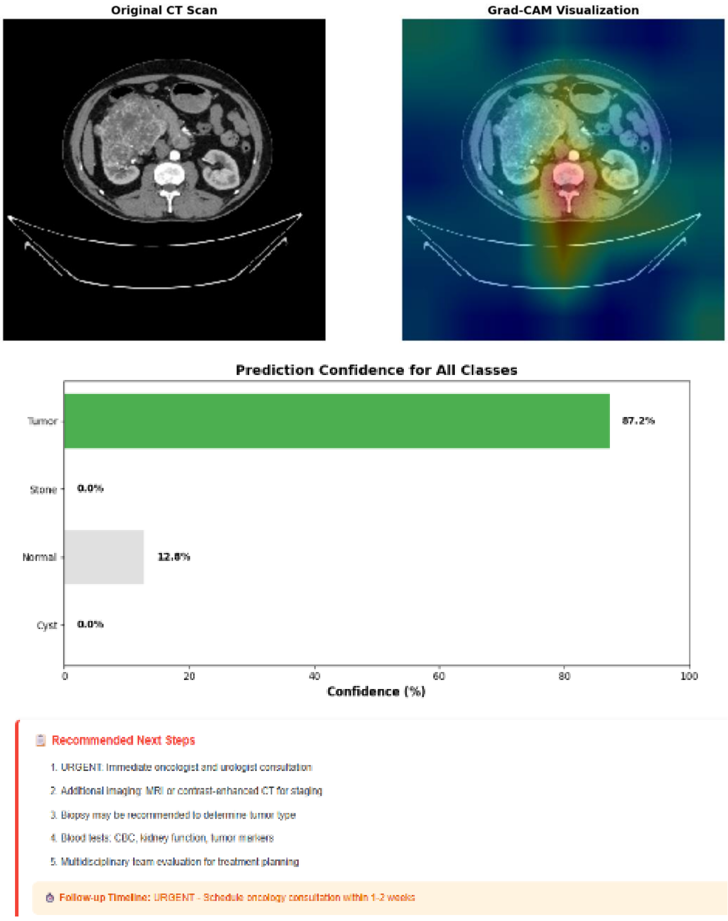


Fig. 7. *Showing what Grad-CAM visualization looks like for a kidney stone prediction*
The highlighted regions, which appear warm match the kidney area where stones are usually found indicating that the model is looking at the right spot for kidney stones.

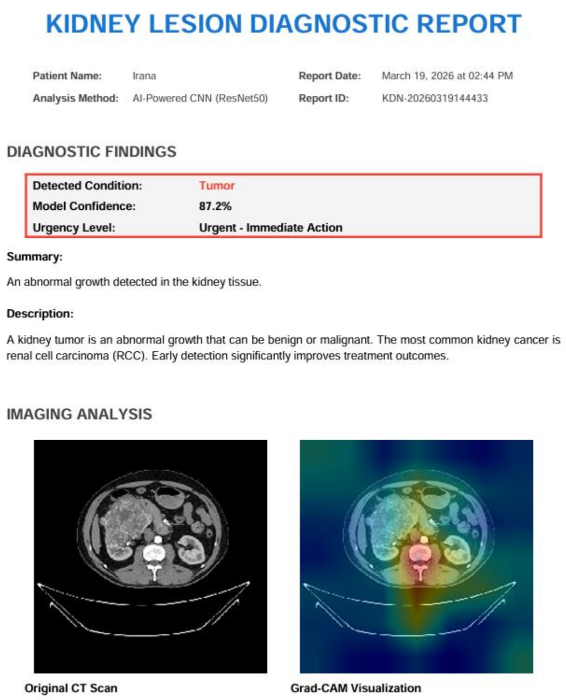
Inspecting the Grad-CAM results I see that ResNet50 always pays attention to the parts of the body. For instance ResNet50 highlights the collecting system when it looks at stones and cysts and it looks at the outer kidney area, which is called the cortex, when it looks at tumors. The Grad-CAM results for ResNet50 show the things that doctors who look at pictures of the body called radiologists usually look for. This is really important. It means that ResNet50 is actually learning things that're useful for doctors like what ResNet50 looks for in pictures of the body rather than just looking at things that do not mean anything or at the background of the pictures.



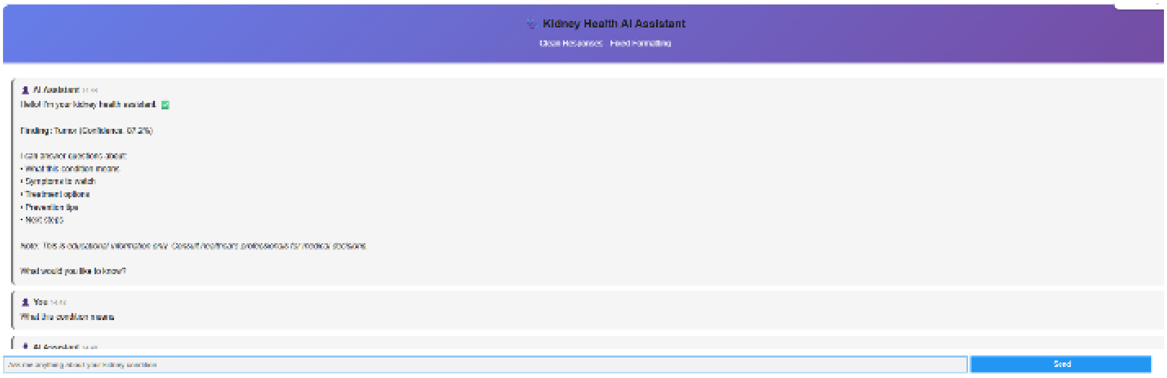
(a) Uploaded CT scan.



(b) Predicted class and confidence.



(c) Report



(d) AI clinical assistant output.

Fig. 8. Clinical prediction interface for a Tumor case (87.2% confidence).

The Stone case in Fig. 8 shows how the system actually works. The model predicts Tumor with a high confidence of 87.2%. However, it is still labelled as Moderate Risk, because even when the model is very sure a doctor should always look at it before any treatment decisions are made.

The AI assistant then provides suggestions based on treatment for kidney stones. It seems sure, about its prediction indicating that the chances are:

Normal: 12.8%, Cyst: 0.0%, Stone: 0.0%, Tumor: 87.2%. The model is pretty confident. Still the system suggests a proper medical check. The kidney stone diagnosis is not something to take lightly.

D. Comparison of models

Reference	Model	Accuracy	Dataset	Explainability
Subedi et al. [5]	Vision Transformer	99.64%	CT Kidney (Kaggle)	No
Pimpalkar et al. [10]	InceptionV3	99.96%	CT Kidney (Kaggle)	No
Alzu'bi et al. [9]	2D-CNN	97.70%	Custom (KAUH)	No
Mehedi et al. [7]	VGG16 + Seg.	99.48%	CT Kidney (Kaggle)	No
Ahmed et al. [12]	VGG16 + Grad-CAM	97.41%	KUB X-ray	Yes
Proposed System	**ResNet50 + Grad-CAM + AI Assistant**	**81.04%**	**CT Kidney (Kaggle)**	**Yes (Full Suite)**

Table V. Comparison of this method with other existing models.

Some previous studies say that they got high accuracy above 99% but we need to be careful when we look at these results. Most of these studies used the Kaggle dataset and did not use any other external validation to check their results. Sometimes the way they split the data might have allowed the training data and the testing data to overlap. So just because a model gets accuracy it does not mean it will work well with new data from different hospitals or different scanners.

In our study we got an accuracy of 81.04%, which's a more realistic and reliable result under the given conditions. What is more important is that our study does more than just classification. Our study combines prediction with explainability, scores how confident we are, provides a clinical assistant, and automates reporting.

The study that is, closest to ours is the one done by Ahmed et al. [12], who also used Grad-CAM with VGG16. However, they used X-ray images and not CT scans, they also did not include a way to help doctors to make decisions based on how confident they are or give them any clinical recommendations.

VIII. DISCUSSION

A. Why ResNet50 Won

The ResNet50 model works better than the rest of the models because of its architecture as compared to the remaining ones. The special connections in ResNet50 help fix a problem that happens when a network gets too complicated. This problem is called the vanishing gradient problem which allows the network to go deeper (up to 50 layers) without losing important learning signals.

Because of this depth, the model can learn features at different levels. The model first starts to learn basic things like edges then it learns to see complicated things like shapes and structures.

This is really important when we are trying to tell the difference between kidney cysts, tumors and stones, where small differences in texture, density, and shape help make out the difference between them clearly.

ResNet50 also performed well in terms of precision by scoring about 0.865. This matters a lot in hospitals and medical settings since we cannot afford to tell someone they are sick when they are actually fine.

B. Why EfficientNetB0 Did Not Work

The performance of EfficientNetB0 was very low at first, but on looking at it more closely, it makes sense. EfficientNetB0 is trained for RGB (colored images) natural images. Hence, when you use it for grayscale (black and white) CT scans images which are quite different in terms of structure and information, the pre-trained features do not transfer well.

As a result, the model does not just perform slightly worse, it failed to learn useful patterns altogether. This shows that just because EfficientNetB0 does well on ImageNet it does not mean it will do well on medical images. For jobs, like this EfficientNetB0 needs to be tested and checked with the kind of data it will be used for, not just with natural image data.

C. Clinical Positioning

This system is made to help doctors to make decisions and for patients to get a faster review of their condition. However, it does not aim to replace the judgement of doctors rather act as an help. The Grad-CAM pictures help doctors understand why the system made a prediction. This way doctors can check the prediction before they do anything and also flags wrong predictions.

The clinical assistant part of the system helps doctors who are not specialists in a particular field. These doctors often need to check for kidney problems before a specialist doctor is available to see the patient. The system gives these doctors the information they need. The system tells doctors what a certain condition usually involves what symptoms they should look out for and what they should do next.
This is especially helpful in areas where it's hard to find specialist doctors and have limited resources but high demand of patients.

D. Limitations

There are some limitations of the work that we need to think about.
First, the testing for this work was done using a dataset from one place so we do not know how well the model will work with data from other hospitals or scanners or in different situations.
Second, the system only looks at one image at a time and does not consider the other factors of patients such as medical history or lab results or other information that doctors normally use to make a diagnosis.
The model also does not give any information about shape of the kidney lesion or size regarding how big it is, which are important things to know when we are planning a treatment.
Lastly, the AI assistant gives people more knowledge about their condition. This advice is the same for everyone with that condition. It does not give advice that's just, for that one person.

E. Future Work

There are a few things which we can do to make this system better:
We need to test the system in hospitals and with scanners to see how it really works.
We should add more details in the system to help it know more about the patient, like their age, their weight and lab results so we can have a deeper knowledge about their condition.
Which can help the system make more accurate predictions.The system has to be trained on data, from hospitals, while keeping the patient's information private.

IX.CONCLUSION

This paper presents the design, implementation, and evaluation of a complete Kidney Lesion Classification System using CNNs with transfer learning. The system focuses on four classes in CT images: Normal, Cyst, Stone, and Tumor.

A comparison of four models on the dataset showed that ResNet50 did the best with an accuracy of 81.04%. The final decision factors showed that it had a precision of 0.865, a recall of 0.810 and an F1-score of 0.819.

On the other hand, EfficientNetB0 performed very poorly, it had an accuracy of 20.23%, which is close to random guessing.This indicates that models that work well on pictures do not always do well with medical images. You need to test models, on the medical data to be sure.

What makes this system different from previous systems is not just its accuracy.It also does a lot of important things combining it all in one pipeline. Grad-CAM helps explain how it makes decisions.There is an assistant that can help not trained doctors and pateints to figure out what to do next. It can also make reports automatically making it easier to understand.

Each part of this system helps solve a problem that doctors face. The model has explainability segments which helps doctors understand what the system is thinking. The risk level's part points out when and how much the system is not sure about something.

This system needs better improvement before it can be used in medical field with full confidence. Most importantly, it needs to know more about the patients it is looking at. In addition, it needs to be tested on data from different hospitals and scanners, focusing more on patient-related information, adding features like lesion segmentation. However, as a proof, the system shows a clear and approach to building an AI tool which is not only accurate but also useful and understandable in medical field.

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